# The Heat Is On: Causal Impacts of Wildfires on Conflict Probability and Intensity in Indonesia


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## QUESTION AND SIGNIFICANCE

**Does wildfire exposure causally increase the probability and intensity of local conflict in Indonesia?**

Indonesia's peatlands are among the most fire-prone landscapes on Earth, with large-scale burning events occurring regularly since the 1990s. Fires destroy agricultural livelihoods, displace communities, and degrade natural resources that rural populations depend on for survival. At the same time, Indonesia has a well-documented history of violent conflict tied to land disputes, resource scarcity, and displacement of indigenous communities by corporate agricultural expansion.

Despite this, the causal link between wildfire shocks and conflict remains understudied. Existing research documents co-occurrence of fires and social unrest, but stops short of establishing causality — leaving a critical gap for policymakers designing climate adaptation and conflict-prevention strategies. This study addresses that gap by exploiting spatiotemporal variation in wildfire exposure across Indonesian districts to estimate its effect on conflict probability and intensity, using ACLED conflict data and satellite-derived fire records.

## LITERATURE AND CONTEXT

**Climate Shocks and Conflict**

Burke et al. (2009), Dell et al. (2012), and Hsiang et al. (2013) collectively establish that climatic shocks — including temperature increases and extreme weather events — significantly raise civil conflict incidence, reduce economic output, and destabilize political institutions across diverse global contexts.

**Resource Scarcity & Violence**

Homer-Dixon (1994) argues that environmental scarcity generates conflict through livelihood disruption and group grievance — a framework directly applicable to rural Indonesia, where agricultural dependence is high and customary land rights are frequently contested by corporate concession holders.

**Indonesia-Specific Context**

Fires in Indonesia are driven by El Niño-induced drought, illegal land clearing for palm oil, and drainage of carbon-rich peatlands. Burgess et al. (2012) and Balboni et al. (2025) show that political incentives and weak property rights enforcement make fire risk in the region a deeply institutional phenomenon.

**Research Gap**

While the literature addresses climate-conflict links and Indonesian fire dynamics, no study directly estimates *causal* effects of wildfire exposure on conflict outcomes at the district level in Indonesia.

## DATA AND METHODS

| Wildfire Data | Conflict Data |
|---|---|
| NASA VIIRS MODIS satellite detections | ACLED georeferenced conflict events |
| Nov 2000 – Feb 2026 | 2015 – 2026 |
| Point-level fire detections → aggregated to district-week | Weekly events → aggregated to district-week |
| Confidence filter: high/nominal only | Event types: battles, riots, Violence Against Civilians, protests |

**Econometric Strategy**

$$Conflict_{it} = \beta_1 \times Fires_{it} + \beta_2 \times Wellbeing_{it} + \alpha_i + \gamma_t + \epsilon_{it}$$

$$Conflict_{it} = \log(\textit{number of conflict events in district and week} - \textit{year})$$
$$Fires_{it} = \log(1 + \textit{Fire Radiative Power}) \textit{ in district } i \textit{ in week} - \textit{year } t$$
$$Wellbeing_{it} = \textit{Human Development Index in province and year}$$
$$\alpha_i = \textit{district fixed effects}$$
$$\gamma_t = \textit{week} - \textit{year fixed effects}$$
$$\epsilon_{it} = \textit{error term}$$

$\beta_1$ is the coefficient of interest, identifying the causal effect of wildfire exposure on conflict probability and intensity.

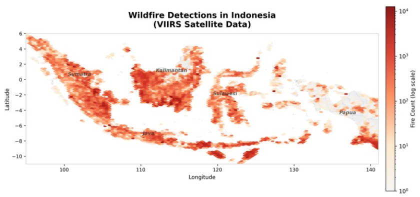
**Limitations**

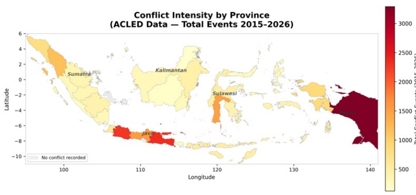
We assume that fire timing is exogenous or as good as random conditional on fixed effects

However, unobserved shocks could bias results.

Future work using an instrumental variables approach will mitigate this limitation.

## FINDINGS




| Lag | Coefficient | P-value | Interpretation |
|---|---|---|---|
| Week 0 | -0.010 | 0.631 | Not significant |
| **Week 4** | **-0.041** | **0.005** | **Significant negative effect** |
| **Week 8** | **-0.047** | **0.011** | **Significant negative effect** |
| Week 12 | -0.0106 | 0.618 | Not significant |
| Week 16 | -0.0101 | 0.461 | Not significant |

Contemporaneous wildfire exposure (log FRP) shows no significant effect on conflict events (β = −0.010, p = 0.631). Most lag coefficients are also statistically insignificant, suggesting fires do not systematically trigger conflict in the short run.

Statistically significant negative coefficients at lags 4 and 8 weeks suggest that fire events may *reduce* reported conflict activity in the weeks following a fire — possibly reflecting displacement of populations, restricted movement, or disruption to normal social activities in fire-affected areas.

**Contrary to standard climate-conflict narratives, wildfires appear to suppress rather than trigger conflict in the short run.**

## FUTURE RESEARCH

- **Instrumental Variable Approach** — Implement wind patterns as an instrument for wildfire exposure, exploiting exogenous variation in fire smoke dispersal to strengthen causal identification
- **Cross-national generalizability** — Extending this framework to other fire-prone, conflict-affected regions (e.g., the Amazon basin, Central Africa) would test whether the Indonesia findings reflect a broader pattern linking climate shocks to local violence
- **Mechanism identification** — Subsequent analysis will disentangle whether conflict is driven primarily by crop loss, displacement, or competition over remaining resources
- **Heterogeneity by land tenure** — Examining conflict responses across formal vs. informal land tenure regions, particularly in palm oil and pulpwood concession areas, will clarify the role of institutional context
- **Policy evaluation** — Future research will exploit Indonesia's Peatland Restoration Agency (BRG) interventions as natural experiments to test whether fire-prevention policies also reduce conflict risk
- **Long-run effects** — This study captures short-run conflict responses; longitudinal analysis tracking communities over 5–10 years post-fire would reveal whether displacement and resource loss generate persistent grievances

## REFERENCES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